

Functora

Hi-end quality software development.
Reliable. Functional. Pure.

About

I am pseudonymous software developer Functora. I do specialize in a software development with Haskell, Rust and Nix languages since 2019. I work with passion and create the best software using the best technologies. My areas of expertise and interest are Bitcoin, Lightning Network, sound money, trading, free markets, privacy and sovereignty. My primary technical goals are type-level guarantees of software correctness and safety.

Skills

Haskell is my first programming language. I do have advanced level of Haskell including:

- GHC.Generics, generic-lens and SYB for efficient data manipulation.
- GHCJS compiler, GHC WASM backend and JSaddle EDSL.
- Presistent and Esqueleto typed SQL libraries.
- Yesod enterprise web framework.
- Miso frontend web framework.

Rust is my second programming language, but nowadays I often prefer it over Haskell due to its superior performance and better development experience. I do have good level of Rust including:

- Chumsky parser combinators.
- Diesel typed SQL library.
- Dioxus cross-platform apps framework.

I am also qualified in other useful areas:

- Nix and NixOS - Strong medium level. Nix is the best tool for deterministic builds, tests and development environments. I am using NixOS daily.
- Bitcoin and Lightning Network - Advanced level as application developer (not as protocol developer). Bitcoin is the greatest achievement of the new millennium, which brings financial freedom and sovereignty back to the people.
- PostgreSQL and SQLite - Strong medium level. I am not database expert, but I am using everything what software developer should use to manipulate data storage. Queries, transactions, joins, locks.
- Docker and Swarm - Advanced close to expert level. I am using Docker and Swarm for development, builds and production. Docker is a very handy tool for MacOS and Linux compatibility.

Code

- cryptonote - A serverless, cross-platform app that allows users to create encrypted or plain-text notes directly on their device or browser. It uses strong authenticated encryption (AES-GCM or ChaCha20-Poly1305) with Argon2id key derivation, embeds note content in shareable URLs or QR codes, packs attachments into offline archives, and ensures no data leaves the device unless explicitly shared. The source code is available on github.

- [functora-css](#) - A classless, JS-free, dependency-free, configuration-free, mobile-first, responsive, lightweight, semantic, pure CSS framework with light and dark themes and a collapsible navbar.
- [functora-tagged](#) - Lightweight, macro-free newtypes with refinement and derived traits.
- [functora-cfg](#) - A Rust library that merges configuration values from multiple sources into a single typed value.
- [currency-converter](#) - An app for converting currencies, generating financial documents, and sharing them via links or QR codes. It includes optional client-side encryption. Built using Miso and GHCJS, the source code is available on [github](#). Mobile version is also available on [Google Play](#).
- [lightning-verifier](#) - An app for offline verification of [Lightning Network](#) invoices and preimages, and sharing them via links or QR codes. It includes optional client-side encryption. Built using Miso and GHCJS, the source code is available on [github](#).

Blog

- [Bitcoin Seed Security](#)
September 18, 2025
- [Offline Lightning Wallet](#)
October 23, 2020
- [Sovereign Web Introduction](#)
October 15, 2020

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